Smt. Chandaben Mohanbhai Patel Institute of Computer Applications

MCA Semester – II

CAUE510 Python Web Framework

Question Bank

Unit-1

1. What is a web framework, and what are its key components?
2. What is Django, and what are its main features?
3. Why is Python commonly used for Django development? Explain the relationship between Django and Python.
4. How does Django incorporate core Python concepts?
5. Explain Django's philosophy of "Batteries Included Framework" with examples.
6. What is the MVT (Model-View-Template) architecture in Django? How does it compare to the MVC (Model-View-Controller) framework?
7. Why is it necessary to create a virtual environment for Django projects?
8. What are the steps to install Django?
9. Explain the structure of a Django project and its key files.
10. What is a Django app, and how does it differ from a Django project?

Unit-2

11. What is a URL, and how does Django handle URLs?
12. What are the different types of URLs in Django? Explain with examples.
13. What is the django.urls module, and what functionalities does it provide, explain each with examples.
14. Explain any two functions from django.urls with examples.
15. What is a view in Django, and how does it work?
16. What are the modules provided by django.http, and what are their uses? Explain any two.
17. What is the request object in Django? Where does the request originate from?
18. Explain the difference between render() and redirect() methods with examples.
19. What is a class-based view (CBV) in Django? How does it differ from function-based views?
20. What is middleware in Django? How does it work?
21. Explain Django’s template system and its syntax with examples.
22. What are template tags and filters in Django? Provide examples.
23. How can you extend templates in Django using the template inheritance feature?
24. How does Django handle static files and media files?

Unit-3,4

25. How do you define a One-to-Many relationship in Django models? Give an example.
26. What are model managers in Django? How can you create a custom manager?
27. How do you create and apply migrations in Django models?
28. Write a Django form class for enrolling students in a course.
29. What is a ModelForm in Django? How does it simplify form creation?
30. How do you validate form input in Django? Explain with examples.
31. Compare CharField, TextField, and SlugField in Django models.
32. What is the difference between ForeignKey, OneToOneField, and ManyToManyField?
33. How does Django handle form submission and CSRF protection?
34. Discuss the security risks related to Django forms and models. How can you prevent them?
35. What are Django Forms? Why are they used in web applications?
36. Explain the difference between Django Forms and ModelForms with examples.
37. How does Django handle form validation? Explain the role of is_valid() and clean() methods.
38. What are Django widgets, and how do they affect form fields? Provide examples.
39. You are building a course enrollment system where students can register for multiple courses using a single form. How would you implement this in Django?
40. Suppose you need to create a form where users can dynamically add or remove multiple instructor experiences (like previous jobs). How would you handle it using Django forms?
41. A course registration form requires a phone number field that must be 10 digits long. How would you enforce this validation?